

Real Clifford Algebra, Triality, and Spinor Gravity in Signature (4,4)

A reproducible exact, geometric, and numerical study

Reproducible exact-real implementation

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Abstract

This dissertation gives an exact-real construction of the Clifford algebra $\mathbf{Cl}(4,4)$, its spin group, the two real half-spin modules, split-octonion multiplication, and split-real triality. The full Clifford algebra is realized by eight integral 16 by 16 matrices and is proved to have rank 256. Its even part preserves two rank-eight eigenspaces of the volume element. Direct commutant and intertwiner calculations prove that these spaces are irreducible and inequivalent as real $\mathbf{Spin}(4,4)$ modules.

An eight-dimensional split composition algebra is independently constructed from Zorn data in an orthogonal basis of signature $(4,4)$. Its multiplication is proved unital, alternative, and norm-composing. Passing to the para-product produces a cyclic invariant trilinear form. The associated related-triple Lie algebra is computed from 512 linear constraints on 84 parameters. It has dimension 28; all three projections are isomorphisms, their eight-dimensional representations are irreducible and pairwise inequivalent, and exact outer operators generate a six-element permutation group.

Four pure-Rust SUNDIALS CVODE studies accompany the exact work. A 24-state transport problem evolves the vector and both half-spin states through one noncommuting Lie-algebra path and preserves all four triality invariants. An 18-state real homogeneous spinor cosmology reconstructs a CPL background and is checked against analytic dilution, density, potential, and Friedmann identities.

The exact representation is then placed on a curved split-signature eight-manifold. A diagonal vielbein determines the curved metric, canonical Levi-Civita spin connection, and rank-16 real spinor bundle. A third 18-state study couples a commuting classical real spinor directly to the Einstein equations without a scalar field or cosmological constant. Finally, the canonical connection is replaced by a flat, metric-compatible, torsionful Weitzenböck connection. Exact Python and Wolfram calculations prove the contortion relation, the teleparallel boundary identity, and equality of the homogeneous Hermitian Dirac operators. An independently implemented fourth CVODE study reproduces the canonical state history byte-for-byte while emitting torsion and boundary diagnostics. Every result is generated from hash-pinned source, independently checked where practical, and reproduced byte-for-byte.

1 1. Scope and claims

The subject is the split real form in eight dimensions. All vector spaces, matrices, fields, and published numerical outputs are real. The principal claims are:

1. $\mathbf{Cl}(4,4)$ is represented faithfully on a real 16-dimensional module.
2. Restriction to the even algebra and to $\mathbf{Spin}(4,4)$ gives two inequivalent real half-spin modules of dimension eight.
3. A split-octonion multiplication table of signature $(4,4)$ follows from an independently defined Zorn model and satisfies the composition law.
4. The vector module and the two half-spin modules are permuted by an outer six-element triality action preserving one trilinear form.
5. The exact tensors can drive reproducible real ODE studies without changing the audited CVODE engine.
6. The real rank-16 module defines a spinor bundle on an explicit curved $(4,4)$ manifold, with canonical Levi-Civita and inertial Weitzenböck spin connections related by contortion.

7. A nonlinear classical-spinor potential supplies dust-like and negative-pressure homogeneous effective-fluid terms in both the Einstein and teleparallel formulations.

The work does not claim a classification of cosmological observations, a microscopic derivation of dark energy, perturbative stability, dimensional reduction to observed $(3,1)$ spacetime, or a particle generation mechanism. The terms "dust-like" and "negative-pressure" describe the homogeneous model; they are not observational identifications of dark matter or dark energy. Historical notebooks are treated as provenance and motivation, never as proof.

2 2. Real Clifford algebra in split signature

Let V be a real vector space with ordered orthogonal basis

$$(e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8)$$

and bilinear form

$$\eta = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

The Clifford algebra is the unital associative algebra generated by V with

$$xy + yx = 2\eta(x, y)1.$$

In particular, the first four basis generators square to $+1$ and the last four square to -1 . The normalization factor two is essential; omitting it is inconsistent with the generator squares.

The standard real classification gives

$$\text{Cl}(4, 4) \simeq \text{Mat}_{16}(\mathbb{R}),$$

and

$$\text{Cl}^0(4, 4) \simeq \text{Mat}_8(\mathbb{R}) \oplus \text{Mat}_8(\mathbb{R}).$$

Thus the full algebra has one irreducible real module of dimension 16, while the even algebra has two inequivalent irreducible real modules of dimension eight. This distinction resolves the most important representation-theoretic ambiguity in the historical material.

3 3. Canonical integral matrix construction

Define three real 2 by 2 matrices

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy

$$P^2 = 1, \quad N^2 = -1, \quad G^2 = 1,$$

and anticommute pairwise. On the fourth tensor power of the real two-space, set

$$\gamma_k^+ = G^{\otimes(k-1)} \otimes P \otimes 1^{\otimes(4-k)},$$

$$\gamma_k^- = G^{\otimes(k-1)} \otimes N \otimes 1^{\otimes(4-k)}, \quad k = 1, 2, 3, 4.$$

The ordered list

$$(\gamma_1^+, \gamma_2^+, \gamma_3^+, \gamma_4^+, \gamma_1^-, \gamma_2^-, \gamma_3^-, \gamma_4^-)$$

consists of integral 16 by 16 matrices and obeys

$$\gamma_a \gamma_b + \gamma_b \gamma_a = 2\eta_{ab} 1_{16}.$$

The 256 ordered monomials

$$\gamma_{i_1} \gamma_{i_2} \cdots \gamma_{i_k}, \quad i_1 < i_2 < \cdots < i_k,$$

including the empty product, have exact matrix rank 256. They therefore span the full matrix algebra and prove faithfulness without relying on a dimension count alone.

4 4. Volume element and half-spin modules

The ordered volume element is

$$\omega = \gamma_1 \gamma_2 \cdots \gamma_8 = G^{\otimes 4}.$$

It satisfies

$$\omega^2 = 1, \quad \omega \gamma_a = -\gamma_a \omega.$$

The exact projectors

$$P_+ = \frac{1 + \omega}{2}, \quad P_- = \frac{1 - \omega}{2}$$

both have rank eight. Odd Clifford elements exchange their images, while even elements preserve them. The spin generators are

$$\Sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b] = \frac{1}{2}\gamma_a\gamma_b \quad (a < b).$$

They satisfy

$$[\Sigma_{ab}, \gamma_c] = \eta_{bc}\gamma_a - \eta_{ac}\gamma_b$$

and the exact $\mathfrak{so}(4,4)$ commutation relations. Restricting the 28 generators to the images of $P+$ and $P-$ gives two eight-dimensional modules.

Three independent tests determine their representation type:

- each even-algebra image has rank 64, the full endomorphism dimension of an eight-space;
- the commutant of each spin action has dimension one;
- the space of intertwiners between the two actions has dimension zero.

The first two statements prove irreducibility over the real numbers. The third proves inequivalence for the fixed Lie-algebra labeling.

5 5. Finite-dimensional representation classification

The Lie algebra $\mathfrak{so}(4,4)$ is the split real algebra of Dynkin type D4. Its four fundamental weights may be labeled so that

- `lambda_1` gives the eight-dimensional vector module V ;
- `lambda_2` gives the 28-dimensional adjoint module;
- `lambda_3` and `lambda_4` give the two eight-dimensional half-spin modules $S+$ and $S-$.

Finite-dimensional irreducible algebraic modules of the simply connected split group are indexed by dominant integral highest weights

$$\lambda = n_1\lambda_1 + n_2\lambda_2 + n_3\lambda_3 + n_4\lambda_4, \quad n_j \in \mathbb{Z}_{\geq 0}.$$

The diagram automorphism group fixes the central weight `lambda_2` and permutes the three outer weights `lambda_1`, `lambda_3`, and `lambda_4`. Consequently triality acts on every highest-weight family by permuting the three outer coefficients. The spacetime-relevant fundamental triple is

$$(V, S_+, S_-).$$

This statement is more precise than saying that the three spaces are simply the same. They are inequivalent modules for a fixed labeling; an outer automorphism changes the labeling and carries one action to another.

6 6. Split-octonion algebra from Zorn data

Write an element as Zorn data

$$X = (a, u, v, b),$$

where $\mathbf{a}, \mathbf{b}$ are real scalars and $\mathbf{u}, \mathbf{v}$ are real three-vectors. Define

$$\begin{aligned} (a, u, v, b)(c, x, w, d) = & (ac + u \cdot w, \\ & ax + du - v \times w, \\ & cv + bw + u \times x, \\ & v \cdot x + bd). \end{aligned}$$

Conjugation is

$$\overline{(a, u, v, b)} = (b, -u, -v, a),$$

and the norm is

$$n(X) = ab - u \cdot v.$$

The orthogonal coordinates used by the implementation are

$$\begin{aligned} a &= x_0 + x_4, & b &= x_0 - x_4, \\ u &= (x_1, x_2, x_3) + (x_5, x_6, x_7), \\ v &= -(x_1, x_2, x_3) + (x_5, x_6, x_7). \end{aligned}$$

They give

$$n(x) = x_0^2 + x_1^2 + x_2^2 + x_3^2 - x_4^2 - x_5^2 - x_6^2 - x_7^2.$$

Exact symbolic verification proves

$$n(xy) = n(x)n(y),$$

the left and right alternative laws, flexibility, conjugation reversal, and the quadratic identity

$$x\bar{x} = n(x)1.$$

In the chosen basis the ordinary multiplication tensor has 64 nonzero integral entries and rank eight in each tensor mode.

7 7. Para-product and invariant trilinear form

Define the para-product

$$x \star y = \bar{x} \bar{y}.$$

It obeys the same norm-composition law. The trilinear form

$$T(x, y, z) = \langle x \star y, z \rangle$$

is cyclic:

$$T(x, y, z) = T(y, z, x) = T(z, x, y).$$

This cyclic form is the clean real algebraic carrier of triality. It avoids choosing one preferred spinor and avoids mistaking a single isometry for the full three-way symmetry.

8 8. Related triples and split-real triality

The related-triple Lie algebra is

$$\mathfrak{tri}(\mathbb{O}_s) = \{(A, B, C) \in \mathfrak{so}(4, 4)^3 : A(x \star y) = Bx \star y + x \star Cy\}.$$

Each skew endomorphism has 28 parameters, so the initial space has 84 parameters. Evaluating the defining equation on all 64 basis pairs gives 512 scalar linear equations. The exact constraint matrix has rank 56 and nullity

1. Each of the three projections from the nullspace to $\mathfrak{so}(4, 4)$ has rank 28.

The three induced eight-dimensional representations have commutant dimensions

$$(1, 1, 1)$$

and pairwise intertwiner dimensions

$$(0, 0, 0).$$

They are therefore irreducible and pairwise inequivalent. Direct intertwiner calculations identify the second and third projections with the two canonical half-spin modules from Section 4.

9 9. Outer six-element action

Cyclic permutation of a related triple defines an order-three automorphism:

$$r(A, B, C) = (B, C, A).$$

Conjugation defines a reflection that fixes the vector position and exchanges the spin positions with the required coordinate conjugations. Its exact matrix has order two. The two operators satisfy

$$r^3 = s^2 = 1, \quad srs = r^{-1},$$

and generate six distinct automorphisms. Because the order-three operator permutes three inequivalent modules, it cannot be induced by conjugation inside the connected spin group. This is the outer triality action.

10 10. Clifford and split-octonion compatibility

Left multiplication by the eight split-octonion basis vectors gives chiral blocks. Combining each block with its conjugate block yields eight real 16 by 16 matrices satisfying the same Clifford relations as Section 3. The space of intertwiners between this multiplication-induced representation and the canonical tensor-product representation has dimension one. A primitive exact intertwiner has rank 16.

This calculation establishes compatibility among three independently defined objects:

1. the tensor-product Clifford representation;
2. the Zorn split-octonion multiplication;
3. the related-triple triality representations.

11 11. Numerical study I: 24-state triality transport

Let

$$y(t) = (v(t), s_+(t), s_-(t)) \in V \oplus S_+ \oplus S_-.$$

Choose one time-dependent Lie-algebra path

$$\begin{aligned} X(t) = & \frac{1}{5}E_{01} + \left(\frac{3}{20} + \frac{t}{40}\right)E_{14} \\ & + \left(-\frac{1}{10} + \frac{t}{50}\right)E_{45} + \frac{t(4-t)}{80}E_{26}. \end{aligned}$$

The selected generators do not all commute and include both compact and split directions. The transport equations are

$$\dot{v} = \rho_v(X)v, \quad \dot{s}_+ = \rho_+(X)s_+, \quad \dot{s}_- = \rho_-(X)s_-.$$

All three initial states equal the unit basis vector. The exact identities

$$\langle v, v \rangle, \quad \langle s_+, s_+ \rangle, \quad \langle s_-, s_- \rangle, \quad T(s_+, s_-, v)$$

are invariants of the ODE.

The application uses pure-Rust SUNDIALS 7.8.0 CVODE with BDF, Newton iteration, a dense linear solver, relative tolerance **1e-11**, componentwise absolute tolerance **1e-13**, and maximum step 0.02. Over $0 \leq t \leq 4$, it writes 41 samples. The verified run used 237 steps and 251 right-hand-side calls. The largest measured invariant drift was less than **3.1e-12**. A second release run produced byte-identical CSV and JSON.

12 12. Numerical study II: real homogeneous spinor cosmology

Use e-fold time $N=\ln(a)$ and the 18-state real vector

$$y = (\tau, \rho, \psi_0, \dots, \psi_{15}),$$

where

$$\tau = H_0(t - t_0), \quad S = \psi^T \psi.$$

The background parameters are

$$\Omega_{m0} = 0.305, \quad \Omega_{r0} = 0.00009, \quad \Omega_{\psi0} = 0.69491,$$

$$w_0 = -0.861, \quad w_a = -0.60, \quad w(a) = w_0 + w_a(1 - a).$$

The reconstructed potential is

$$U(S) = \Omega_{\psi0} S^{1+w_0+w_a} \exp \left[-3w_a \left(1 - S^{-1/3} \right) \right].$$

Choose a verified real skew Clifford matrix J with $J^2=-1$. The equations are

$$\frac{d\tau}{dN} = \frac{1}{E},$$

$$\frac{d\rho}{dN} = -3[1 + w(e^N)]\rho,$$

$$\frac{d\psi}{dN} = -\frac{3}{2}\psi + \frac{U,S}{E}J\psi,$$

with

$$E^2 = \Omega_{r0}e^{-4N} + \Omega_{m0}e^{-3N} + \rho.$$

Skewness of J gives

$$\frac{dS}{dN} = -3S, \quad S = e^{-3N}.$$

The continuity equation then gives the analytic CPL density, and substitution gives $\rho=U(S)$. CVODE integrates independently from the present epoch to $N=-4$ and $N=1$, then joins the branches into 1,201 ordered samples. The verified run used 711 steps and 783 right-hand-side calls. Maximum relative errors were

Identity	Maximum relative error
condensate dilution	6.57e-9
analytic density	8.67e-10
potential reconstruction	7.71e-9
Friedmann closure	3.08e-11

Repeated release runs produced byte-identical outputs.

The calculation establishes a homogeneous background realization. It does not establish perturbative stability. The region with $w<-1$ particularly requires an independent analysis of fluctuations and admissible initial data.

13 13. Curved Spin(4,4) geometry and the canonical spin connection

Let M be an oriented, time-oriented, spin eight-manifold with tangent metric

$$\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

Use coordinates

$$(x_0, x_1, x_2, x_3, t, y_1, y_2, y_3)$$

and choose a positive scale factor $\mathbf{a}(\mathbf{t})$. The explicit curved metric is

$$ds^2 = a(t)^2 \left[(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 - (dy^1)^2 - (dy^2)^2 - (dy^3)^2 \right] - dt^2.$$

The selected diagonal coframe and its inverse are

$$e_\mu{}^a = \text{diag}(a, a, a, a, 1, a, a, a),$$

$$e_a{}^\mu = \text{diag}(a^{-1}, a^{-1}, a^{-1}, a^{-1}, 1, a^{-1}, a^{-1}, a^{-1}).$$

They satisfy the defining vielbein relation

$$g_{\mu\nu} = e_\mu{}^a \eta_{ab} e_\nu{}^b.$$

The associated real spinor bundle is

$$\mathcal{S} = P_{\text{Spin}(4,4)}(M) \times_\rho \Delta_{\mathbb{R}}, \quad \Delta_{\mathbb{R}} = \Delta_+ \oplus \Delta_-.$$

Thus a field $\mathbf{psi}$ is a section of a rank-16 real bundle whose two chiral summands have rank eight. A global construction requires the spin-structure assumption; the displayed frame is a local gauge choice.

Define

$$H = \frac{\dot{a}}{a}.$$

The Levi-Civita Christoffel symbols are

$$\Gamma^\rho{}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

For transverse indices I in $\{0, 1, 2, 3, 5, 6, 7\}$, the nonzero components are

$$\Gamma^4{}_{II} = \eta_{II} a \dot{a}, \quad \Gamma^I{}_{4I} = \Gamma^I{}_{I4} = H.$$

The vielbein postulate

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0$$

gives the canonical spin connection

$$\omega_\mu{}^a{}_b = e_b{}^\nu (\Gamma^\rho{}_{\mu\nu} e_\rho{}^a - \partial_\mu e_\nu{}^a).$$

Its nonzero lowered components are

$$\omega_{II4} = \eta_{II}\dot{a}, \quad \omega_{I4I} = -\eta_{II}\dot{a}.$$

The spinor derivative is

$$D_\mu^{LC}\psi = \partial_\mu\psi + \frac{1}{8}\omega_{\mu ab}[\gamma^a, \gamma^b]\psi,$$

where both ordered tangent indices are summed. Exact contraction gives

$$\gamma^\mu D_\mu^{LC}\psi = \gamma^4 \left(\partial_t + \frac{7}{2}H \right) \psi$$

for a homogeneous spinor. Independent Python and Wolfram derivations verify all components of the vielbein postulate, connection antisymmetry, chirality preservation, and this contracted operator.

The geometry has signature (4,4). Its constant-time slices have signature (4,3), so it is not ordinary one-time cosmology.

14. Numerical study III: coupled Einstein-spinor dynamics

Let

$$C = \gamma_1^+ \gamma_2^+ \gamma_3^+ \gamma_4^+, \quad \bar{\psi} = \psi^\top C, \quad S = \bar{\psi}\psi.$$

The symmetric bilinear $\mathbf{C}$ has signature (8,8) and is invariant under the local spin action. The classical commuting-spinor action is

$$I = \int_M d^8x \sqrt{|g|} \left[\frac{R}{2\kappa_8} + \frac{1}{2} \left(\bar{\psi} \gamma^\mu D_\mu^{LC} \psi - (D_\mu^{LC} \bar{\psi}) \gamma^\mu \psi \right) - V(S) \right].$$

No scalar field and no cosmological constant occur. Choose

$$V(S) = mS + \lambda S^q, \quad \kappa_8 = 21, \quad m = \frac{1}{20}, \quad \lambda = \frac{19}{20}, \quad q = \frac{1}{5}.$$

The homogeneous density and pressure are

$$\rho = V(S), \quad p = SV'(S) - V(S) = (q-1)\lambda S^q.$$

The reduced Einstein equations are

$$21H^2 = \kappa_8 \rho,$$

$$6\dot{H} + 21H^2 = -\kappa_8 p.$$

The homogeneous spinor equation is

$$\dot{\psi} = -\frac{7}{2}H\psi - V'(S)\gamma^4\psi.$$

Skew-adjointness of γ^4 with respect to $\mathbb{C}$ implies

$$\dot{S} = -7HS, \quad S(a) = a^{-7}$$

for $S(1)=1$. Consequently

$$\rho(a) = ma^{-7} + \lambda a^{-7q}.$$

The first term has zero homogeneous pressure and dilutes as seven-dimensional dust. The second has equation of state

$$w_q = q - 1 = -\frac{4}{5}$$

and dilutes more slowly.

The state

$$y = (a, H, \psi_0, \dots, \psi_{15}) \in \mathbb{R}^{18}$$

is integrated from $t=0$ to $t=-0.2$ and $t=1.5$, then joined into 171 ordered samples. The canonical run uses relative tolerance $1\text{e-}11$, absolute tolerance $1\text{e-}13$, maximum step 0.002, 1,372 internal steps, and 1,486 right-hand-side evaluations. Maximum relative errors are below $7.5\text{e-}9$ for condensate dilution, density reconstruction, and Friedmann closure. A refined run uses $1\text{e-}12$, $1\text{e-}14$, and maximum step 0.001.

At the normalized epoch,

$$S = \rho = H = 1, \quad p = w = -0.76,$$

with dust-like fraction 0.05 and negative-pressure fraction 0.95. The acceleration changes sign at

$$t = -0.1380052933032450, \quad a = 0.8631436165767085.$$

These labels describe homogeneous effective-fluid behavior. The calculation does not establish clustering, perturbative stability, dimensional reduction, or agreement with observational data.

15 15. Weitzenböck spin connection and teleparallel equivalence

For a coframe $\mathbf{e}$ and a flat inertial tangent connection $\omega^a{}_{W\mu b}$, define

$$\Gamma^\rho{}_{W\mu\nu} = e_a{}^\rho \left(\partial_\mu e_\nu{}^a + \omega^a{}_{W\mu b} e_\nu{}^b \right).$$

In the selected diagonal proper-frame gauge,

$$\omega^a{}_{W\mu b} = 0, \quad \Gamma^\rho{}_{W\mu\nu} = e_a{}^\rho \partial_\mu e_\nu{}^a.$$

The only nonzero affine coefficients are

$$\Gamma^I{}_{W4I} = H.$$

The connection is metric-compatible and curvature-free. With

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{W\mu\nu} - \Gamma^\rho{}_{W\nu\mu},$$

its nonzero torsion components are

$$T^I{}_{4I} = H, \quad T^I{}_{I4} = -H.$$

The torsion trace used by the spinor equation is

$$au_\mu = T^\nu{}_{\mu\nu}, \quad au_4 = 7H, \quad au_I = 0.$$

Define contortion by

$$K^\rho{}_{\mu\nu} = \Gamma^\rho{}_{W\mu\nu} - \Gamma^\rho{}_{LC\mu\nu}.$$

Then

$$\Gamma_W = \Gamma_{LC} + K,$$

and the corresponding spin lifts obey

$$\Omega_W = \Omega_{LC} + K_{\text{spin}} = 0$$

in the selected gauge. The zero representative is gauge-dependent. Under a local Lorentz transformation it becomes a generally nonzero pure-gauge connection; it must not be interpreted as an invariant absence of gravity.

The torsion scalar is

$$\mathbb{T} = \frac{1}{4}T^\rho{}_{\mu\nu}T_\rho{}^{\mu\nu} + \frac{1}{2}T^\rho{}_{\mu\nu}T^{\nu\mu}{}_\rho - \tau_\mu\tau^\mu = 42H^2.$$

The Levi-Civita scalar curvature and teleparallel boundary term are

$$R_{LC} = 14\dot{H} + 56H^2,$$

$$B = 14\dot{H} + 98H^2,$$

so

$$\boxed{R_{LC} = -\mathbb{T} + B.}$$

The TEGR gravitational action is therefore

$$I_g = -\frac{1}{2\kappa_8} \int_M d^8x e \mathbb{T},$$

which differs from the Einstein-Hilbert action by a boundary divergence.

For the symmetric spinor action, integration by parts in the torsionful geometry contributes half the torsion trace. The Euler-Lagrange equation is

$$\gamma^\mu \left(D_\mu^W + \frac{1}{2}\tau_\mu \right) \psi - V'(S)\psi = 0.$$

Since $\Omega_W=0$ in the selected gauge,

$$\gamma^\mu \left(D_\mu^W + \frac{1}{2}\tau_\mu \right) \psi = \gamma^4 \left(\partial_t + \frac{7}{2}H \right) \psi.$$

Thus the homogeneous Hermitian Weitzenböck and Levi-Civita Dirac operators agree although the underlying connections have different curvature and torsion.

16. Numerical study IV: independent teleparallel spinor dynamics

The Phase 6 application owns its matrices, state types, right-hand side, and CVODE setup. It generates `C` and `gamma^4` directly from the exact fixtures and depends only on the pinned solver crates. It does not call the canonical Einstein-spinor application.

The teleparallel field equations reduce to the same 18 first-order equations only after the boundary and Hermitian Dirac identities are proved. The independent canonical run emits, at every sample,

$$\dot{H}, \quad \tau_4, \quad \mathbb{T}, \quad R_{LC}, \quad B, \quad R_{LC} + \mathbb{T} - B,$$

and the difference between the two homogeneous Dirac coefficients.

All 171 serialized times and state components agree exactly with the Levi-Civita run. The maximum boundary-identity residual is $2.842170943040401\text{e-}14$, and the homogeneous Dirac-coefficient residual is exactly zero. Repeated release runs are byte-identical. The refined run differs from the canonical state by at most $6.310776406656671\text{e-}10$ in normalized components.

This agreement is a test of teleparallel equivalence for the specified action and homogeneous ansatz. The torsion scalar rewrites the gravitational sector; it is not counted as an additional dark fluid. Nonminimal terms such as $F(S)$, T , $G(S)$, B , $f(T)$, or $f(T,B)$ would define different theories and are not included.

17 17. Reproducibility architecture

The repository separates evidence by ownership boundary:

- `wolfram/` contains exact symbolic definitions;
- `artifacts/exact/` contains hash-linked machine-readable tensors;
- `vendor/sundials_rs` is a read-only Git submodule;
- `studies/` contains application-owned ODE behavior;
- `artifacts/` contains canonical numerical results;
- `notebooks/` contains generated Mathematica and Jupyter presentations;
- `provenance/` contains complete commands for each published artifact.

The exact fixtures are generated twice and compared by SHA-256. Independent Python and Wolfram programs reconstruct their defining identities. Each numerical study is run twice in release mode; semantic checkers recompute invariants from CSV and require byte-identical replay. The two gravitational studies also use tighter-tolerance convergence runs. The generated Mathematica notebook executes 20 input cells with nine fail-fast checks. PowerShell and Git Bash entry points are both tested from anonymous recursive clones.

18 18. Scientific limitations and future work

The exact algebraic conclusions are finite computations over integers and rational numbers. The numerical conclusions are consistency results for two specified ODE systems. Neither category supports claims not encoded in its hypotheses.

Future work should address:

1. global group integration and explicit central-character bookkeeping;
2. orbit and stabilizer classification for null as well as non-null triality triples;

3. perturbations of the homogeneous spinor background;
4. data likelihoods with published covariance rather than central values;
5. geometric field equations coupling the split-signature internal structure to a physical $(3,1)$ spacetime model and a justified compactification;
6. independent numerical comparison on additional platforms.
7. perturbations and stability for both the Levi-Civita and teleparallel spinor backgrounds;
8. nonminimal torsion-spinor models treated as distinct theories rather than inferred from TEGR.

19 19. Conclusion

The project establishes, in one reproducible real framework, the exact relationship among $\mathbf{Cl}(4,4)$, its 16-dimensional module, the two real half-spin modules, split-octonion multiplication, and triality. The representation distinction is explicit: the full Clifford module is irreducible, its restriction to the spin group splits, and the three eight-dimensional fundamental modules are inequivalent until permuted by an outer automorphism.

The four numerical studies demonstrate how these exact tensors enter application-owned CVODE models without changing the audited solver. The curved construction distinguishes the tangent metric, coordinate metric, vielbein, Levi-Civita spin connection, and inertial Weitzenböck connection. Exact contortion and boundary identities explain why two independently implemented gravitational studies produce the same homogeneous state history.

The nonlinear spinor potential separates into dust-like and negative-pressure homogeneous terms without introducing a scalar field or cosmological constant. Those are precise properties of the stated classical model, not observational detections. The invariants, analytic comparisons, hashes, convergence tests, gauge qualifications, and scientific limitations are all part of the published result.

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